



Wun-Yi (Wing) Chen

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About me

Astrophysics Ph.D. candidate specializing in multidimensional radiation hydrodynamics of supernova shock breakout and circumstellar interaction. My research focuses on multigroup radiation transport, large-scale HPC simulations, and multiwavelength observables. Currently conducting research at HITS through the DAAD–NSTC program.

Projects

DAAD–NSTC Research Fellowship at HITS, Heidelberg | 28 Feb 2026 - Current

Hosted by Prof. Friedrich K. Röpke at the Heidelberg Institute for Theoretical Studies (HITS).

- Conduct multidimensional radiation-hydrodynamic simulations of supernova shock breakout and circumstellar interaction.
- Study observable multiwavelength signatures of red-supergiant explosions interacting with confined CSM.
- Collaborate on connecting radiation-hydrodynamic simulations with stellar-evolution models and observations.

NTU & ASIAA Supernova Shock Breakout Research | 1 Sep 2020 - Current

Supervisor: Dr. Ke-Jung Chen.

- Develop multidimensional multigroup radiation-hydrodynamic simulations of supernova shock breakout using CASTRO/AMReX.
- Investigate shock breakout from red and blue supergiants interacting with winds and confined circumstellar material.
- Model angle-dependent and multiwavelength observables for comparison with early-time supernova observations.
- Perform large-scale simulations and post-processing on HPC systems using C++, Fortran, Python, and MPI-based workflows.

Education & Training

Ph.D. Candidate in Astrophysics (Direct-Track Program) | National Taiwan University, Graduate Institute of Astrophysics | 1 Sep 2021 - Current | Taipei, Taiwan

Field(s) of study: Physics | **Thesis:** Investigating the First Light of Supernovae: Radiation-Hydrodynamic Simulations of Shock Breakout

Graduate Study in Astrophysics | National Taiwan University, Graduate Institute of Astrophysics | 1 Sep 2020 - 30 Jun 2021 | Taipei, Taiwan

Field(s) of study: Physics | **Final grade:** GPA 4.17/4.3

Bachelor's Degree, Double Major in Physics and Business Administration | National Taiwan University | 1 Sep 2015 - 30 Jun 2020 | Taipei, Taiwan

Field(s) of study: Physics , Management and administration

Exchange Program | TUM School of Management | 1 Sep 2017 - 31 Aug 2018 | Munich, Germany

Field(s) of study: Management and administration

Publications

Multiwavelength Signatures of Supernova Shock Breakout from Red Supergiants in Two Dimensions

2026. Wun-Yi Chen, Ke-Jung Chen, Keiichi Maeda, Masaomi Ono, Po-Sheng Ou, and Friedrich K. Röpke. ApJ 999 56.

Multidimensional Radiation Hydrodynamics Simulations of Supernova 1987A Shock Breakout

2024. Wun-Yi Chen, Ke-Jung Chen, Masaomi Ono. ApJ 976 147.

Observable Signatures of Supernova Shock Breakout in Confined Circumstellar Medium

2026. (2026. Submitted / preprint) Wun-Yi Chen, Ke-Jung Chen, Keiichi Maeda, and Friedrich K. Röpke.

Conferences & Seminars

Astronomische Gesellschaft Annual Meeting. Contributed Oral Presentation

"Multiwavelength Signatures of Supernova Shock Breakout" | 7 Sep 2026 - 11 Sep 2026 | Garching, Germany

EAMA 11. Contributed Oral Presentation "Supernovae Shock Breakout with Confined-Shell CSM in Two Dimensions" | 8 Dec 2025 - 12 Dec 2025 | Japan

SNRIII. Flash Talk "Multidimensional Radiation Hydrodynamics Simulations of Supernova 1987A Shock Breakout" | 9 Jun 2024 - 15 Jun 2024 | Crete, Greece

Taiwanese Theoretical Astrophysics Workshop II. Local Organizing Committee (LOC) | 23 Nov 2022 - 25 Nov 2022 | Taipei, Taiwan

Other Conferences | 1 Sep 2019 - 30 Sep 2022 | Taipei, Taiwan

- *21st Heidelberg Summer School: Cosmic Multiplicity: Evolution and Fate of Binary Stars*
- *Les Houches Summer School: Interstellar Shocks School*
- Additional presentations: 7 contributed talks and 9 posters at national and international conferences and workshops.

Honours and Awards

DAAD/NSTC Sandwich Scholarship | NSTC/DAAD | 28 Feb 2026

NSTC Subsidy for Domestic Graduate Students to Attend International Academic Conferences | NSTC | 8 Dec 2025

NCTS Student Outstanding Paper Award | NCTS | 10 Sep 2025

DAAD-NSTC Summer Institute Program | DAAD-NSTC | 1 Jul 2025

Chung Hwa Rotary Education Foundation Scholarship | Chung Hwa Rotary Education Foundation | 1 Jul 2025

NSTC Subsidy for Domestic Graduate Students to Attend International Academic Conferences | NSTC | 9 Jun 2024

Financial Support | Interstellar Shocks School of Les Houches | 12 Jun 2022

Top Admission | Graduate Institute of Astrophysics | 1 Sep 2020

Work experience

University teaching assistant | National Taiwan University, Department of Physics & Graduate Institute of Astrophysics | 1 Sep 2020 - Current | Taipei, Taiwan

- Radiation Hydrodynamics
- Classical Mechanics
- Modern Physics Lab
- General Physics I and II

Skills

Computational | C++ | Fortran | Python | Bash

Scientific Computing | CASTRO/AMReX | HPC | Parallel Computing | MPI

Tools | Linux | Git | scientific visualization and data analysis

Language Skills

Mother tongue(s): **Chinese**

	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	C2	C2	C1	C1	C1
German	B1	B2	B1	B1	B1

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user